

CIS 20 – MICROCOMPUTER ASSEMBLY LANGUAGE BERKELEY CITY COLLEGE

Tentative Course Schedule Fall 2026

Instructor Ying Liu

Course details

- Credit hours: 4
- Lec/Lab: MW 6:30 to 9:15 PM

Questions or Comments

- **Classroom:** Room 323, Berkeley City College
- **Office Hours:** After class or by appointment (Zoom)
- **Email:** yingliu@peralta.edu

Prerequisite:

- CIS 006 or CIS 025 or CIS 036A

Student Learning Outcomes

Upon completion of this class, students will be able to do the following:

1. Describe computer hardware architecture and the organization and functions of key components.
2. Explain binary logic and digital circuits, the structure of the CPU, and how computers perform arithmetic and logic operations.
3. Explain and/or demonstrate how computers execute instructions using the fetch-execute cycle and the structure of low-level computer instructions.
4. Write simple programs for microcontrollers using assembly language.

Details of Topics Covered

The course covers the following topics:

- Computer System: CPU, ALU, cache structure, memory, and basic functions of operating system such as process scheduling, and memory management
- Number systems: Binary numbers, representation of signed and unsigned numbers, floating point systems and arithmetic operations
- Machine-level representation of programs: Arithmetic and logic operation, control, procedures, and arrays.
- The Organization and Function of the Central Processing Unit and Instruction Sets:

Characteristics of instruction sets, addressing modes, instruction sets, registers, ALU, control unit, instruction cycle, and pipelining

- Memory hierarchy and virtual memory
- Parallel Organization: Instruction-level parallelism, multiple processor organization, cache coherence, multicore computers, and cloud computing
- x86/x86-64 Machine-Level Programming: Basic instructions, registers, addressing, program control, procedures, and the relationship between high-level code and machine-level execution.

Notes

This course explores what happens beneath high-level programming languages: how computers represent information, execute instructions, organize memory, and interact with software. We will use assembly language as one tool for understanding these concepts. No prior assembly-language experience is expected.

It is very important that you attend lectures, read the suggested material, and do the assignments.

Required Textbook

- *Computer Systems: A Programmer's Perspective*, 3rd ed., Randal E. Bryant and David R. O'Hallaron

Grading

- Assignments/Homework: **50%**
- Midterm Exam: **25%**
- Final Exam: **25%**
- Extra Credit: **up to 3 percentage points**

Extra Credit: Limited extra-credit opportunities may be offered during the semester. Extra credit may increase a student's final course grade by a maximum of **3 percentage points**. Extra credit cannot replace missing assignments, the midterm, or the final exam.

Final Grade Scale

A: 90–100%
B: 80–89.9%
C: 70–79.9%
D: 60–69.9%
F: Below 60%

Important Dates

Verify these dates with the Admissions and Records Office

- **August 24 (Monday):** Fall instruction begins
- **August 30 (Sunday):** Last day to add without an instructor permission number
- **September 7 (Monday):** Labor Day — no classes
- **September 15 (Tuesday):** Last day to drop without a “W”

- **October 13 (Tuesday):** Professional Day — no classes
- **November 11 (Wednesday):** Veterans Day — no classes
- **November 20 (Friday):** Last day to withdraw with a “W”
- **November 26–29:** Thanksgiving holiday
- **December 12 (Saturday):** Last day to file for Pass/No Pass
- **December 13–19:** Final examinations
- **December 19 (Saturday):** Fall semester ends

Please note that you must select your grading option (A—F or CR/NC) before the college specified deadline. The instructor cannot change the grading option.

Disabilities

If you have a documented disability and wish to discuss academic accommodations, please contact me as soon as possible. You can also contact the Disability Services and Programs for Students for assistance.

Homework and Lab Assignments

All assignments must be typed and submitted electronically through Canvas unless otherwise specified. Handwritten assignments will not be accepted. Diagrams, calculations, and mathematical notation should be presented clearly in a readable digital format.

Assignments provide hands-on practice with course concepts and programming. Some assignments may be challenging and open-ended, so students are encouraged to start early and use lab time to ask questions and work with instructor support.

You may seek help and discuss course concepts with classmates and other resources, but submitted work must represent your own understanding. You must be prepared to explain your programs, calculations, and answers.

Class participation

Please attend all lectures and labs and be prompt. Please complete any assigned reading before you come to class. Obviously, you will not understand it fully, but it will help you anticipate what topics would be covered in the lecture. Write a brief summary of what you expect during the lecture and be prepared to ask questions.

I wish to make this course as accessible as possible to all students. Please let me know in advance anything that may affect any aspect of the course assignments or participation and will improve your learning experience.

You can earn extra credit by participating in class discussions, helping other students, taking extra-credit quizzes, and submitting all assignments in a timely manner. Please note that you need to be outstanding in these activities to get full credit. Extra-credit is assigned by the instructor and not discussed with the students.

Academic Integrity and Use of AI Tools

All submitted work must represent your own understanding. You may discuss course concepts and general approaches with classmates, but assignments must be completed individually unless collaboration is explicitly permitted.

AI tools such as ChatGPT, Claude, Gemini, GitHub Copilot, and similar tools may be used to help explain concepts, debug code, or explore alternative approaches unless an assignment specifically prohibits their use. However, you are responsible for understanding everything you submit. You must be able to explain your code, calculations, reasoning, and answers.

Copying solutions from another student, the Internet, solution repositories, or AI-generated output without understanding and independently completing the work is not acceptable. Specific assignments, quizzes, the midterm, and the final exam may prohibit outside or AI assistance entirely.

Academic dishonesty will be handled according to Berkeley City College and Peralta Community College District policies.

Classroom Expectations

- Computers and mobile devices should be used for course-related activities during class. Please minimize activities that may distract you or other students.
- Please keep phones on silent during class. If you need to make or receive a call, please step outside.
- You are encouraged to discuss course concepts and lab work with other students. However, submitted work must be completed independently unless collaboration is specifically permitted.

Tentative weekly schedule

- **Weeks 1–2:** Introduction to computer systems; information representation; binary and hexadecimal; signed and unsigned integers
- **Weeks 3–5:** Integer arithmetic; data representation in memory; pointers, arrays, strings; floating-point representation
- **Weeks 6–8:** Machine-level representation and basic x86/x86-64; registers; memory addressing; arithmetic and logic operations; program control; procedures and the stack
- **Week 9:** Machine-level programming practice; assignment review; midterm preparation
- **Week 10:** Midterm Exam; introduction to processor architecture
- **Week 11:** Processor architecture; instruction execution; pipelining; program performance and optimization
- **Weeks 12–13:** Memory hierarchy; locality; cache memory; cache mapping and performance; cache-friendly programming
- **Weeks 14–15:** Virtual memory; physical and virtual addressing; paging; page tables; address translation; TLBs; memory mapping
- **Week 16:** Operating systems, processes, I/O, system calls, exceptions and interrupts; course integration and final review
- **Week 17:** Final Exam